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Root systems

Antoine de Saint Germain

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Theorem 1. The 8-tuple $(\mathbb{Z}, \mathbb{Z}^l, (\mathbb{Z}^l)^*, \mathcal{L}, I, \alpha, \alpha^\vee, s)$ is a root pairing of A_l , where:

1. $\mathcal{L}$ is the dot product on $\mathbb{Z}^l \times (\mathbb{Z}^l)^*$
2. $I = \{\text{signed intervals on the set } \{1, \dots, n\}\} = I^+ \cup I^-$
3. $s : I \rightarrow \text{Perm}(I)$ where for $J \in I$, $s_J = s_{-J}$, $s_J(-K) = -s_J(K)$, defined as $s_J(K) =$

$$\begin{cases} J \cup K & \text{if } J \cap K = \emptyset \wedge J \cup K \in I \\ K & \text{if } J \cap K = \emptyset \wedge J \cup K \notin I \\ K \setminus J & \text{if } J \cap K \neq \emptyset \wedge J \subset K \wedge K \setminus J \in I \\ K & \text{if } J \cap K \neq \emptyset \wedge J \subset K \wedge K \setminus J \notin I \\ -K & \text{if } J \cap K \neq \emptyset \wedge K \subseteq J \wedge J \setminus K = \emptyset \\ -(J \setminus K) & \text{if } J \cap K \neq \emptyset \wedge K \subseteq J \wedge J \setminus K \in I \\ K & \text{if } J \cap K \neq \emptyset \wedge K \subseteq J \wedge J \setminus K \notin I \\ K & \text{if } J \cap K \neq \emptyset \wedge K \not\subseteq J \wedge J \not\subseteq K \end{cases}$$

Equivalently (for use in Lean),

1. $\mathcal{L}$ is the dot product on $\mathbb{Z}^l \times (\mathbb{Z}^l)^*$
2. $I = \{(i, j, \varepsilon) | 1 \leq i \leq j \leq l \in \mathbb{Z}, \varepsilon = \pm 1\}$
3. $\alpha : I \hookrightarrow \mathbb{Z}^l$, $(i, j, \varepsilon) \mapsto \varepsilon \left(\sum_{k=i}^j A_l e_k \right)$ (where A_l is the Cartan Matrix of A_l)
4. $\alpha^\vee : I \hookrightarrow (\mathbb{Z}^l)^*$, $(i, j, \varepsilon) \mapsto \varepsilon \left(\sum_{k=i}^j e_k^\top \right)$
5. $s : I \rightarrow \text{Perm}(I)$ where $s(i, j, 1) = s(i, j, -1)$, $s(i, j, \varepsilon)(k, l, 1) = (p, q, \eta)$ means that $s(i, j, \varepsilon)(k, l, -1) = (p, q, -\eta)$, and

$$s(i, j, 1)(k, l, 1) = \begin{cases} (k, l, -1) & \text{if } (i, j) = (k, l) \\ (l+1, j, -1) & \text{if } i = k, j > l \\ (j+1, l, 1) & \text{if } i = k, j < l \\ (i, k-1, -1) & \text{if } i < k, j = l \\ (k, i-1, 1) & \text{if } i > k, j = l \\ (k, j, 1) & \text{if } i = l+1 \\ (i, l, 1) & \text{if } k = j+1 \\ (k, l, 1) & \text{otherwise} \end{cases}$$

Proof.

□